Partially Ordered Stochastic Conformance Checking: A Review

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1 The Technique

Leemans et al. generalise Earth Movers' Stochastic Conformance (EMSC) from totally ordered to partially ordered traces. The motivation is semantic. When two activities execute concurrently, the order in which they are recorded carries no process information, so any conformance method that depends on it is measuring a product of the process properties rather than the properties themselves. The technique therefore re-bases conformance checking on a different semantic object—the *stochastic partially ordered (PO) language*—and lifts EMSC to that level. Three concrete limitations of standard EMSC are addressed simultaneously: **P1** uncertain event order in logs (events sharing a timestamp), **P2** redundant assignment of probability to interleavings of concurrent behaviour, and **P3** the absence of any formal guarantee on values returned for models containing loops.

1.1 Semantic Objects

A PO trace is a triple $\rho = (E, \prec, l) \in \mathcal{P}$ with finite event set E, strict partial order $\prec\, \subseteq E \times E$, and labelling $l\colon E \to \mathcal{A}$. Its set of linearisations $\mathcal{L}(\rho) \subseteq \mathcal{A}^*$ collects every total order consistent with $\prec$, so one PO trace stands for a family of classical traces. A *stochastic PO language* is a map $L\colon \mathcal{P} \to [0,1]$ with $\sum_\rho L(\rho) = 1$, generalising classical stochastic languages (where every $\prec$ is total). PO traces admit two interpretations: *certain*, where the executor freely picks any linearisation (used for models, addressing **P2**), and *uncertain*, where one linearisation was realised but cannot be recovered from the data (used for logs with imprecise timestamps, addressing **P1**).

1.2 Embedding Models and Logs into the Semantics

The model side is a labelled stochastic Petri net (LSPN) $PN = (P, T, F, \Sigma, \lambda, M_0, w)$ with weight function $w\colon T \to \mathbb{R}^+$ assigning probabilities. The authors require soundness, safeness and *confusion-freeness*: any two transitions t_1, t_2 with $\bullet t_1 \cap \bullet t_2 \neq \emptyset$ and $\bullet t_1 \neq \bullet t_2$ are never jointly enabled. Under these properties the probability of a run depends only on transitions involved in choices; weights on concurrently firing transitions cancel out,

$$PN(r) \;=\; \prod_{t' \in T'} \frac{w(\beta(t'))}{\sum_{t \in T \,\mid\, \bullet t \,=\, \bullet\beta(t')} w(t)}, \tag{1}$$

so the order in which concurrent transitions are interleaved leaves $PN(r)$ unchanged. Algorithm 1 performs a recursive play-out, firing maximal sets of pairwise-independent enabled transitions in parallel at each step. Each run defines a PO trace by removing silent transitions and all places while keeping the partial order over the remaining events; summing $PN(r)$ over runs sharing a PO trace yields L_M(probabilities of traces discovered by the LSPN). Because loops admit infinitely many runs, the play-out is *truncated* after q most-likely plus r random runs, and the residual mass $L_M(\perp)$ is treated as an extra symbolic PO trace. This adresses **P3**. On the log side, each case yields one PO trace by reading $\prec$ off the timestamps—events sharing a timestamp stay incomparable—and empirical frequencies are normalised to L_L under the uncertain interpretation; reliable timestamps allow the certain interpretation instead.

1.3 Distance, EMSC, and Bounds

A trace–trace distance δ (the paper uses normalised Levenshtein) is lifted to a PO-trace distance Δ. Because each side may be certain or uncertain, depending on the interpretation of Model traces(L), four variants arise: Δ^{cc}_{δ} (both certain, minimum over pairs of linearisations), $\Delta^{uu\text{-best}}/\Delta^{uu\text{-worst}}$ (both uncertain), and mixed minimax $\Delta^{uc\text{-best}}/\Delta^{uc\text{-worst}}$. EMSC is then the standard optimal-transport problem

$$\mathrm{emsc}_{\Delta}(L, L') \;=\; 1 \;-\; \min_{R} \sum_{\rho \in \tilde{L}, \rho' \in \tilde{L}'} R(\rho, \rho')\, \Delta(\rho, \rho'), \tag{2}$$

subject to the marginal constraints $L(\rho) = \sum_{\rho'} R(\rho, \rho')$ and $L'(\rho') = \sum_{\rho} R(\rho, \rho')$, but now indexed over PO traces. Δ^{cc} is computed by an A^* search over a four-move alignment (synchronous, substitution, top, bottom), with an admissible heuristic equal to the multiset difference of yet-unaligned labels. When L_M is truncated and/or L_L is uncertain, the true EMSC is bounded: absorbing $L_M(\perp)$ into the symbolic trace $\perp$ with cost 0 (lower bound) or 1 (upper bound) and selecting the appropriate Δ-variant yields an interval $[\mathrm{emsc}_{\uparrow}, \mathrm{emsc}_{\downarrow}]$ that provably contains the true value (Lemmas 4 and 5). The gap is equal to $1 - \sum_{\rho \in \tilde{L}_M} L_M(\rho)$ and shrinks monotonically as q and r grow.

2 Advantages

Semantic alignment. By moving conformance to stochastic PO languages, the verdict measures process structure rather than recording artefacts. The toy example of the paper, where classical EMSC yields 0.689 and PO-EMSC yields 0.955, is symptomatic: 45% of the model's probability mass was forced onto two interleavings of one concurrent pair under the totally ordered view, neither of which the log matched exactly; in the PO view the mass sits on a single concurrent PO trace that both interleavings of the log satisfy.

Statistical economy. A single PO trace represents up to $|\mathcal{L}(\rho)|$ total orders, which can be factorial in the width of $\prec$. The applicability study reports 2000 PO traces that collectively represent up to $2.1 \cdot 10^{11}$ classical traces; the same coverage with classical EMSC would require sampling on the order of 10^{20} traces. The bottleneck shifts from sampling to distance computation.

Formal guarantees for loops and uncertainty. The bound lemmas turn a previously heuristic truncation into a verifiable computation: the user knows the true value lies in $[\mathrm{emsc}_{\uparrow}, \mathrm{emsc}_{\downarrow}]$, and can spend more compute on q and r to tighten it. The same lemmas treat log uncertainty (logs with many equal timestamps such as Sepsis at 30%) symmetrically, returning an honest interval rather than a single number that depends on an arbitrary tie-breaking rule.

Decoupling concurrency weights from choice weights. The confusion-free assumption proves that weights on transitions outside any choice do not affect run probabilities, simplifying modelling and matching the practice of translating standard BPMN with exclusive, parallel, and block-structured inclusive gateways into Petri nets.

3 Disadvantages

Worst-case factorial in concurrency width. Counting $|\mathcal{L}(\rho)|$ and computing Δ^{cc} are factorial in the number of pairwise-concurrent events. The implementation defends itself by returning a baseline of 0 whenever $|\mathcal{L}(\rho)| > 10^5$, which loosens the bound. In logs with wide concurrency—exactly those where the PO perspective adds the most value—this fallback weakens the very guarantee that motivates the method.

Slow convergence of emsc-uc**.** The Sepsis experiment leaves a 0.049 gap with $q = r = 1000$, and the BPIC18-4 case shows bound widths up to 0.824 even after 10^6 truncated PO traces. The paper itself flags $\Delta^{uc\text{-worst}}$ as future work. On the very logs where the uncertain interpretation is needed, the technique often returns an interval too wide to discriminate between competing models.

Unverified structural assumptions. The implementation assumes the input LSPN is confusion-free, sound, and safe, but does not verify these properties. A user supplying a model violating one of them receives a silently incorrect answer instead of an error, which is a deployment hazard rather than a theoretical limitation.

Confusion-freeness and soundness needs to be verified manually. Δ^{cc} While safeness can be guaranteed by the discovery algorithm, there is still a need to verify confusion-freeness and soundness. This is a non-trivial task, especially for large process models. The author mentions that automated checks for these properties still need to be developed, which adds an additional hurdle for practical adoption of the technique.